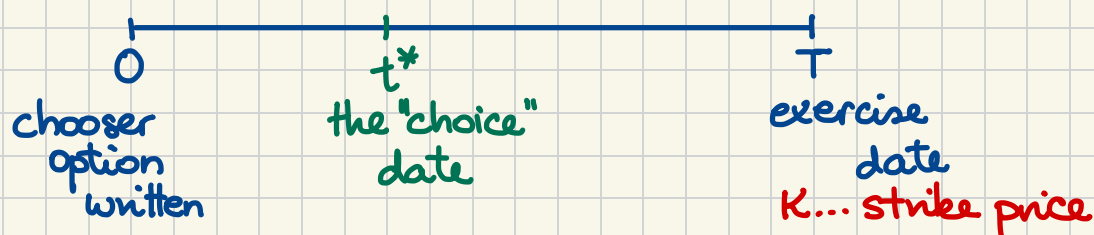


Chooser Options (aka "as you like it" options).



At time t^* , the chooser option's owner decides whether the option becomes a call or a put (always w/ strike K and exercise date T).

Assume that the owner is rational.

Q: What is their criterion for choosing between a call and a put @ time t^* ?

→: Notation.

- $V_{CH}(t, t^*, T)$
 - t : valuation date
 - t^* : choice date
 - T : exercise date
- $V_{C/P}(t, \text{exercise date, strike price})$

Our criterion:

$$V_{CH}(t^*, t^*, T) = \max(\overbrace{V_C(t^*, T, K)}^a, \overbrace{V_P(t^*, T, K)}^b)$$
$$= V_C(t^*, T, K) + \max(0, V_P(t^*, T, K) - V_C(t^*, T, K))$$

|| PUT-CALL PARITY
 $Ke^{-r(T-t^*)} - S(t^*)$

$$= V_C(t^*, T, K) + \left(\underbrace{Ke^{-r(T-t^*)}}_{K^*} - S(t^*) \right)_+$$

The payoff of a European put w/ strike K^* and exercise date t^*

A replicating portfolio for the chooser option:

- a long call w/ strike K and exercise date T
- a long put w/ strike K^* and exercise date t^*

$$\Rightarrow V_{CH}(0, t^*, T) = V_C(0, T, K) + V_P(0, t^*, K^*)$$
$$= V_P(0, T, K) + V_C(0, t^*, K^*)$$

20. Assume the Black-Scholes framework. Consider a stock, and a European call option and a European put option on the stock. The current stock price, call price, and put price are 45.00, 4.45, and 1.90, respectively.

Investor A purchases two calls and one put. Investor B purchases two calls and writes three puts.

The current elasticity of Investor A's portfolio is 5.0. The current delta of Investor B's portfolio is 3.4.

Calculate the current put-option elasticity.

- (A) -0.55
- (B) -1.15
- (C) -8.64
- (D) -13.03
- (E) -27.24

21-24. DELETED

25. Consider a chooser option (also known as an as-you-like-it option) on a nondividend-paying stock. At time 1 its holder will choose whether it becomes a European call option or a European put option, each of which will expire at time 3 with a strike price of \$100. choice date (t^*)

The chooser option price is \$20 at time $t = 0$.

$$V_{CH}(0, 1, 3) = 20$$

$T \dots$ exercise date

The stock price is \$95 at time $t = 0$. Let $C(T)$ denote the price of a European call option at time $t = 0$ on the stock expiring at time T , $T > 0$, with a strike price of \$100.

You are given:

- (i) The risk-free interest rate is 0. $K^* = K$

- (ii) $C(1) = \$4$. $V_C(0, 1, 100) = 4$

Determine $C(3)$. $V_C(0, 3, 100) = ?$

- (A) \$ 9
- (B) \$11
- (C) \$13
- (D) \$15
- (E) \$17

$$V_{CH}(0, 1, 3) = 20 = V_C(0, 3, 100) + V_P(0, 1, K^* = 100)$$

$$= V_C(0, 1, 100) + K^* - S(0)$$

$$= 4 + 100 - 95 = 9$$

$$V_C(0, 3, 100) = 20 - 9 = 11$$

